Practice problems #16

1. Suppose $PA = LU$ where P is a permutation matrix. Explain the steps to solve the equation $Ax = b$.

2. Let $A = \begin{bmatrix} 0 & 0 & 6 \\ 1 & 2 & 3 \\ 0 & 4 & 5 \end{bmatrix}$.

   (a) Give a permutation matrix P so that PA is upper triangular.

   (b) Give two permutation matrices P_1 and P_2 so that P_1AP_2 is lower triangular.

3. Find a 3×3 permutation matrix $P \neq I$ with $P^3 = I$.

4. True or False. Justify.

   (a) Suppose V is a set of all 2×2 real matrices with *nonnegative* entries. With the sum and scalar multiplications defined as usual (that is, elementwise, as we know of), V is a vector space.

   (b) Suppose V is a set of 2×2 symmetric matrices. With the sum and scalar mulitiplications defined as usual (that is, elementwise, as we know of), V is a vector space.

Answers are on the back.

#1: Premultiply P to the equation to get $PAx = Pb$ (that is, permute entries of b) which then becomes $LUx = b', (b' = Pb)$. Solve this equation as usual LU.

#2(a): $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$

#2(b): $P_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad P_2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

#3: $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ or P^T.

#4(a): False. Hint: closed under scalar multiplication? counter-example? Note that to be a vector space, the scalar multiplication thing should be true for *all* scalars.

#4(b): True. Sum of two symmetric matrices symmetric? How about scalar multiple? Is there zero vector? (Is zero matrix symmetric?) Is additive inverse of a symmetric matrix symmetric?) Must show all (10) axioms are satisfied.